

K230 Target Tracking

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K230 and MSPM0 Communication

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K230 and MSPM0 Communication

1. Experimental Prerequisites

This tutorial uses the MSPM0G3519 development board. The corresponding example path is 【14.export\MSPM0-K230\15_k230_nano_tracker】 .

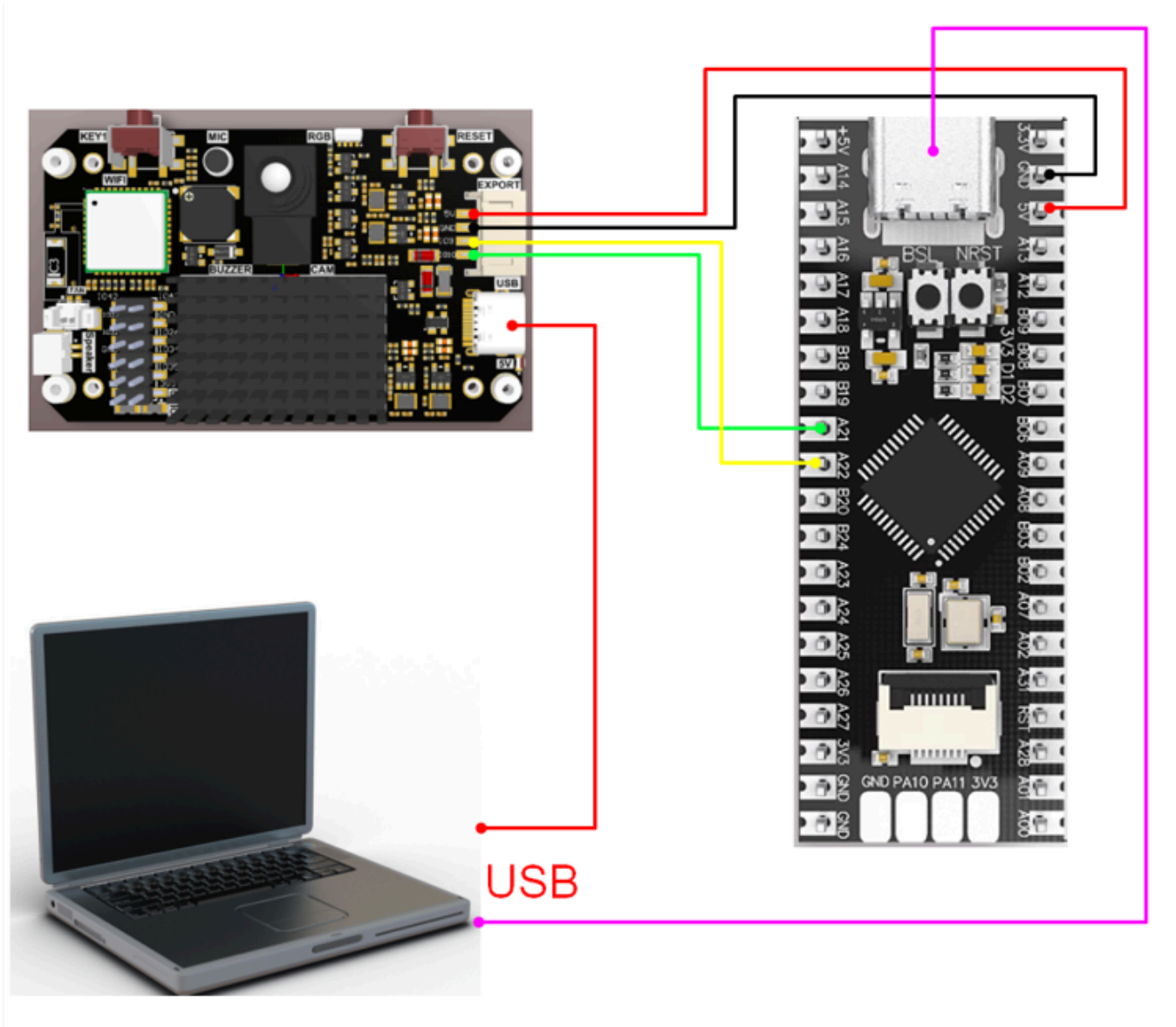
K230 needs to run the 【14.export\CanmvIDE-K230\15.nano_tracker.py】 program to start the experiment. It is recommended to download as an offline running program.

Items needed:

- Windows system computer
- MSPM0G3519 development board
- K230 vision module (with TF card burned with image)
- Two type-C data cables
- Connecting cables

2. Experimental Wiring

K230 Vision Module	MSPM0 Development Board
5V	VCC
GND	GND
TXD(IO9)	A22
RXD(IO10)	A21



3. Main Code Explanation

```
void Pto_Data_Parse(uint8_t *data_buf, uint8_t num)
{
    uint8_t pto_head = data_buf[0];
    uint8_t pto_tail = data_buf[num-1];
    if (!(pto_head == PTO_HEAD && pto_tail == PTO_TAIL))
    {
        sprintf(print_buf, "pto error:pto_head=0x%02x , pto_tail=0x%02x\n", pto_head,
pto_tail);
        uart0_send_string(print_buf);
        return;
    }
    uint8_t data_index = 1;
    uint8_t field_index[PTO_BUF_LEN_MAX] = {0};
    int i = 0;
    int values[PTO_BUF_LEN_MAX] = {0};
    for (i = 1; i < num-1; i++)
    {
        if (data_buf[i] == ',')
```

```

    {
        data_buf[i] = 0;
        field_index[data_index] = i;
        data_index++;
    }
}

for (i = 0; i < data_index; i++)
{
    values[i] = Pto_Char_To_Int((char*)data_buf+field_index[i]+1);
}

uint8_t pto_len = values[0];

if (pto_len != num)
{
    sprintf(print_buf, "pto_len error:%d , data_len:%d\n", pto_len, num);
    uart0_send_string(print_buf);
    return;
}
uint8_t pto_id = values[1];
if (pto_id != PTO_FUNC_ID)
{
    sprintf(print_buf, "pto_id error:%d, func_id:%d\n", pto_id, PTO_FUNC_ID);
    uart0_send_string(print_buf);
    return;
}
int x = values[2];
int y = values[3];
int w = values[4];
int h = values[5];
sprintf(print_buf, "tracker:x:%d, y:%d, w:%d, h:%d\n", x, y, w, h);
uart0_send_string(print_buf);
}

```

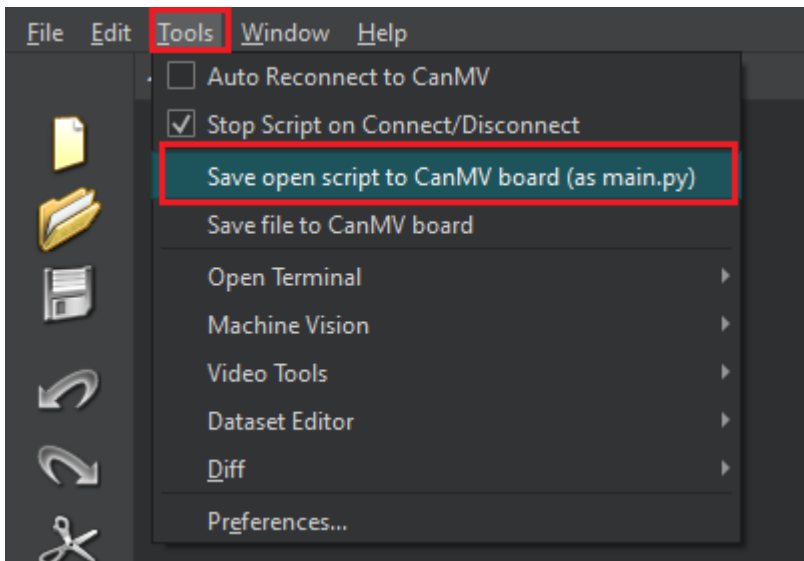
The above function is for parsing K230 data. Only data conforming to the specific protocol can be parsed to get the corresponding data.

Where:

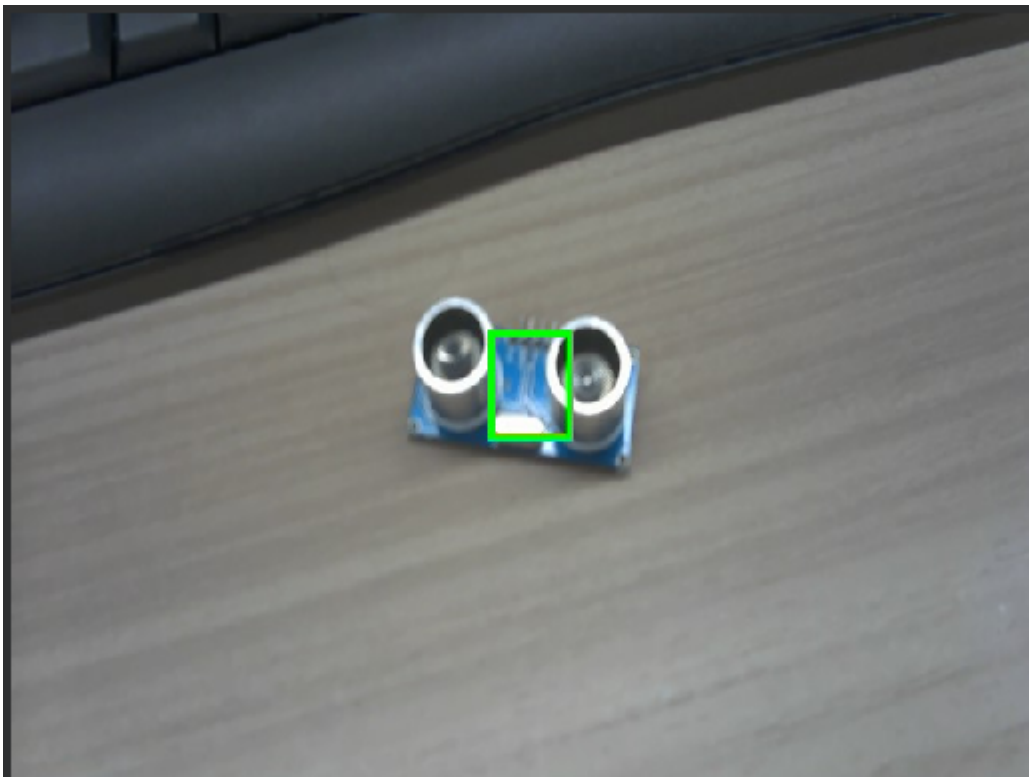
- x: is the x-coordinate of the upper-left corner of the recognized box
- y: is the y-coordinate of the upper-left corner of the recognized box
- w: is the width of the recognized box
- h: is the height of the recognized box

4. Experimental Phenomenon

1. After connecting the cables, the K230 vision module runs offline
Connect K230 to Canmv IDE, open the corresponding program, click 【Save open script to CanMV board (as main.py)】 in the toolbar, then restart K230.



When K230 is powered on, a green box will appear on the screen. Please align the green box with the target to track (ultrasonic sensor is used here as an example). Wait a few seconds. When the green box turns red, it means the target has been detected. At this time, you can move the target object and see the red box following the movement.





Note: Do not move the tracked object too quickly, otherwise tracking may fail.

2. MSPM0 burns the hex file generated by the example.

UniFlash Session ▾ About ? ⚙

Configured Device : Serial Connection > MSPM0G3507(BOOTLOADER) [\[download ccxml\]](#) ●CORTEX_M0P

Program	Select and Load Images
Settings & Utilities	Flash Image(s)
Standalone Command Line	☐ Password Browse ☒ Application Image 1 C:\Users\user\Documents\example.hex Size: 12.81 KB Browse × ☐ Application Image 2 Browse ☐ Application Image 3 Browse
	Available Action(s) - 1 Image Selected Load Image Note: Please power cycle your device before loading images

3. Set the serial assistant interface as shown in the figure

Port
COM25:USB 串行设备

Baud rate 115200

Stop bits 1

Data bits 8

Parity None

Operation  Close

Save Data Clear Data

☐ Hex ☐ DTR

☐ RTS ☐ 自动保存

☐ TimeStamp 1000 ms

4. When the K230 camera image recognizes the target object, the serial assistant will print the information transmitted by K230 to MSPM0.

Where:

- x: is the x-coordinate of the upper-left corner of the recognized box
- y: is the y-coordinate of the upper-left corner of the recognized box
- w: is the width of the recognized box
- h: is the height of the recognized box

As shown in the figure below

```
[2025-04-30 12:13:26.935]# RECV ASCII>
tracker:x:231, y:161, w:34, h:81

[2025-04-30 12:13:26.983]# RECV ASCII>
tracker:x:233, y:164, w:34, h:80

[2025-04-30 12:13:27.140]# RECV ASCII>
tracker:x:234, y:168, w:34, h:80
tracker:x:235, y:172, w:34, h:80
tracker:x:237, y:174, w:34, h:80

[2025-04-30 12:13:27.299]# RECV ASCII>
tracker:x:236, y:176, w:34, h:80
tracker:x:236, y:179, w:34, h:80
tracker:x:237, y:179, w:34, h:80

[2025-04-30 12:13:27.347]# RECV ASCII>
tracker:x:239, y:180, w:34, h:78
```